

ARMAAN GUPTA

☎ 437-332-5557 ✉ a585gupt@uwaterloo.ca  [linkedin.com/in/gupta-armaan](https://www.linkedin.com/in/gupta-armaan)  armaan.world

Education

University of Waterloo

Sept. 2024 – Apr. 2029

Bachelor of Computer Science + Minor in Artificial Intelligence

Waterloo, ON

Technical Skills

Languages: Java, Python, C, C++, C#, SQL, JavaScript, TypeScript, HTML/CSS, Swift, PHP, Ruby on Rails

Frameworks, Tools, & Libraries: React, Node.js, Git, AWS, TensorFlow, PyTorch, Firebase, Angular, Docker, Next.js, npm, Flask, pandas, NumPy, MongoDB, OpenCV, Tailwind, FastAPI, REST API, Vite, Three.js, Figma, Linux, Bash, GCP

Experience

Ford

Jan. 2026 – Present

Software Engineer

Toronto, ON

- Eliminated manual order ingestion paths by designing automated validation and **persistence pipelines**, reducing human intervention by **60%+**.
- Instrumented distributed services with Datadog APM (tracing, metrics, alerts), enabling **low-latency** debugging and proactive performance **regression detection**.

Baba Care

Sept. 2025 – Jan. 2026

Engineer

New York, NY

- Joined as one of the **first engineers**, building core infrastructure that facilitated a **\$4M+ Pre-Seed** led by General Catalyst
- Engineered the **most comprehensive care-center search platform** in the USA, aggregating data across **40,000+** hospitals, nursing homes, and more, generating **10,000+** visits within weeks.
- Engineered a prescription intelligence engine using **SQLAlchemy**, and Supabase that aggregates free, reliable public APIs to identify generic alternatives, ensuring data accuracy while **eliminating licensing costs**.
- Optimized the provider approval workflow, saving some patients over **\$1,200 annually** per recurring refill.

WATonomous

July 2025 – Present

Software Engineer

Waterloo, ON

- Built an Automated Power Outage Notification System, leveraging **cloud-based services** and containerized microservices with Docker, reducing manual reporting delays by **80%**.
- Developed cloud-native APIs and automation scripts to integrate with SLURM and Kubernetes, enabling **parallel execution** of simulation jobs and increasing throughput by **300%**.
- Developed and containerized backend services with Docker, deploying to cloud and **HPC clusters** via Kubernetes.

Environment Canada (ECCC)

May 2025 – Aug. 2025

Software Engineer

Toronto, ON

- Developed a climate reanalysis model pipeline by integrating **ML-based** interpolation techniques using **scikit-learn** and **PyTorch**, improving **spatial resolution** of missing environmental data by **35%**.
- Built an interactive web tool to explore **40+ years** of climate data across **50+** stations and **16 datasets**.

Projects

Assembl3D | *Next.js, Three.js, Gemini Vision, Node.js, Bright Data*

 [repo link](#)

- Built a generative AI platform converting 2D PDF manuals into interactive 3D visualizations, leveraging a **microservices architecture** with Next.js and **React Three Fiber**.
- Engineered a scraping pipeline to harvest product manuals, feeding a multi-modal Gemini vision system to extract **3D spatial geometry** and assembly logic.

PrepPal | *SQL, Python, JS, React, Tailwind, Flask, Google Cloud, OpenAI, Cohere*

 [repo link](#)

- Developed a **full-stack** interview preparation assistant to provide interactive mock interviews and AI-driven feedback.
- Integrated OpenAI and Cohere APIs to formulate personalized GenAI feedback and **real-time response analysis**, improving user preparation by **40%**, resulting in **500+ unique users** within 2 weeks of launch.